

Executive Summary

PASS RATE 28% target 90%	TOTAL TESTS 168 executed	PASSED 48 29% of total	FAILED 120 71% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 28% | Target: 90% | Gap: +62%

The Revenue Management System Kenya backend API testing reveals significant reliability concerns with a 28% pass rate across 168 tests, substantially below the 90% target threshold. The API Automation suite shows 27.5% success rate (22/80 passed) while Backend Smoke Validation achieved 29.5% (26/88 passed), indicating systemic issues across multiple endpoints and service layers. The consistent failure patterns across both test suites suggest fundamental backend stability issues rather than isolated endpoint problems, with the majority of API responses likely experiencing either incorrect data formatting, missing required fields, authentication/authorization failures, or service availability issues that require immediate technical investigation and remediation.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	22	54	29%	FAIL
2	BE Smoke Validation	88	26	62	30%	FAIL
TOTAL		168	48	120	28%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#52	06 Jul	168	28%	120	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	48	28.6%	180ms	Within SLA
Requests Failed	120	71.4%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	6	5%
High	8	7%
Medium	4	3%
Low	2	2%

QUALITY INDICATORS

Indicator	Value	Target
Endpoint Coverage	88%	85%
Execution Rate	100%	100%
Contract Validation	95%	90%
Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate root cause analysis on the top 10 most frequently failing endpoints to identify common failure patterns	Immediate	P1
2	Review backend service logs for error patterns, timeout issues, and database connection problems during test execution periods	3 days	P1
3	Verify API response schemas and data contracts are correctly implemented across all revenue management endpoints	5 days	P2
4	Implement enhanced monitoring and logging for all backend services to capture detailed failure diagnostics	1 week	P2
5	Prioritize fixing authentication/authorization middleware if access control failures are contributing to test failures	1 week	P2
6	Schedule technical deep-dive session with backend development team to review test failure logs and establish remediation timeline	2 weeks	P3

KEY FINDINGS

1	Backend services achieving only 28% overall pass rate against 90% target, indicating critical stability issues
2	API Automation suite showing 58 endpoint failures suggesting widespread response format or data integrity problems
3	Smoke validation failures across 62 tests pointing to fundamental service availability or configuration issues
4	Consistent failure patterns between both test suites indicating systemic backend problems rather than isolated incidents
5	Critical revenue management endpoints experiencing reliability issues that could impact financial operations

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation



29%

